CRIMEVISIONAI

Brainstorming & Idea Prioritization

Folder 1 | Ideation deliverable | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Project concept

CrimeVisionAI is a multimodal forensic evidence analysis and investigation-support prototype. It helps an investigator organize uploaded audio, images and video, extract machine-generated observations, assemble timestamped events, and ask questions over case material. It is decision support—not an autonomous crime solver or a substitute for forensic expertise.

2. Brainstormed problem areas

Idea	Investigation pain point	Value if addressed	Decision
Multimodal evidence intake	Evidence sits in separate files and formats	One case workspace for multiple modalities	Prioritize
Speech transcription and event extraction	Manual listening and note-taking are time-consuming	Searchable transcript and candidate events	Prioritize
Image/video visual analysis	Reviewing many frames is laborious	Object detections, captions and scene cues	Prioritize
Unified chronological timeline	Events from different evidence sources are difficult to compare	Time-oriented review across evidence items	Prioritize, with caveats
Grounded investigator Q&A	Investigators need to locate supporting observations quickly	Faster retrieval from case analysis	Prioritize but validate grounding
Automated legal conclusions / guilt prediction	High risk of unsupported conclusions and misuse	Not appropriate for this prototype	Reject
Full forensic chain-of-custody certification	Requires formal controls, auditability and validated procedures	Important for future production work	Future / out of current scope

3. Prioritization method

Priorities were assessed qualitatively using investigator usefulness, feasibility within a student prototype, dependency on available models, and risk of overclaiming. This is a reasoned project prioritization, not a measured user study.

Priority	Capability	Reason
P0 — Core	Case workspace and evidence upload	Foundation for organizing evidence.
P0 — Core	Audio transcription and timestamp extraction	Converts audio into searchable text and candidate events.
P0 — Core	Image/video visual analysis	Provides visual observations from images and sampled frames.
P1 — Core workflow	Timeline and case-level Q&A	Connects analysis outputs for review; grounding must be tested.
P1 — Quality	Error handling, traceable outputs and clear uncertainty	Needed to avoid presenting model inference as established fact.
P2 — Future	PDF case reports, richer evidence fusion, scalable storage	Useful extensions once core correctness is improved.
P2 — Future	Speaker diarization, audit trail, deployment hardening	Requires dedicated implementation and validation.

4. Prioritization outcome

The chosen minimum viable scope is a local web application that accepts supported evidence, runs modality-specific analysis, stores case/evidence records, exposes analysis through APIs, and supports timeline-oriented review and case Q&A.; Highest-risk follow-up work is grounding/consistency validation, followed by repeatable testing and report generation.

5. Ethical and operational guardrails

Use synthetic or lawfully authorized test evidence. Keep evidence access restricted. Treat object detections, captions, transcriptions and LLM-generated event descriptions as fallible leads. Preserve the original uploaded file, distinguish observation from inference, and require human review before any investigative decision.